

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	25 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

The Human Development Index (HDI) Predictor project aims to develop an intelligent machine learning-based web application that predicts the Human Development Index score and development category of a country using important socio-economic indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per capita.

During brainstorming sessions, the team explored several ideas including HDI score prediction, country development comparison, interactive data visualization, development trend analysis, policy recommendation support, and classification of countries into Very High, High, Medium, and Low human development categories.

After evaluating each idea based on feasibility, implementation complexity, and practical usefulness, the team selected a Linear Regression-based HDI Prediction System integrated with a Flask Web Application as the most suitable solution.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Objective

Bring the project team together to discuss the problem and define the project goals.

Selected Problem Statement

How might we help governments, researchers, students, and policymakers accurately predict the Human Development Index (HDI) score of a country using machine learning techniques and socio-economic indicators, enabling better development planning and decision-making?

Step-2: Brainstorm, Idea Listing and Grouping

Brainstormed Ideas

- HDI Score Prediction using Machine Learning
- Country Development Classification
- Interactive Flask Web Application
- Country Comparison Dashboard
- Data Visualization using Matplotlib & Seaborn
- Feature Importance Analysis
- Development Trend Analysis
- User-Friendly Prediction Interface
- Input Validation
- Model Performance Evaluation
- Download Prediction Report
- Future HDI Forecasting

Grouping

Group A – Machine Learning

- Linear Regression Model
- Model Evaluation
- HDI Prediction
- Feature Selection

Group B – User Interface

- Flask Web Application
- HTML Forms
- Country Selection
- Prediction Result Page

Group C – Data Processing

- Data Cleaning
- Missing Value Handling
- Label Encoding
- Feature Engineering

Group D – Data Visualization

- Heatmaps
- Scatter Plots
- Distribution Plots
- Correlation Matrix
- Strip Plots

Step-3: Idea Prioritization

Prioritized Ideas

Priority	Idea
1	Machine Learning HDI Prediction
2	Flask-Based Prediction System
3	Interactive Data Visualization
4	Country Development Classification
5	Future Enhancement: HDI Trend Forecasting

Prioritization Matrix

High Impact & High Feasibility

- HDI Prediction using Linear Regression
- Flask Web Application
- HDI Category Classification

Medium Impact

- Country Comparison Dashboard
- Interactive Charts

Future Enhancements

- AI-based Policy Recommendation
- Time Series HDI Forecasting
- Global Development Analytics Dashboard

Selected Solution

Machine Learning-based Human Development Index (HDI) Prediction System using Linear Regression integrated with a Flask Web Application.